

Stability Assay and Lot-Disposition Exhibit

Admitted October 7, 2025

United States District Court for the Western District of North Carolina Blue Ember Biologics, LLC v. Granite Heron Logistics, Inc. Civil Action No. SYN-WDNC-24-CV-0520

Blue Ember's quality custodian assembled this exhibit from deviation BE-QA-DEV-2024-0912, the confirmatory assay summary, sample accountability, and the September 16, 2024 disposition for validation lot BE214-VL-2409A. The materials were maintained in the program quality system before this lawsuit's expert analysis.

The lot was manufactured to support a process-validation campaign, not commercial patient distribution. Its release record incorporated a continuous minus-eighty to minus-sixty transport condition. Receiving at 15:26 on September 12 triggered quarantine after personnel reviewed monitor CTX9-44721 and observed the fifty-four-minute excursion. Seals remained intact, and the receiving signature did not release the material.

The exhibit separates sampled biological activity from fitness for the intended validation purpose. Blue Ember does not assert that every vial was chemically inactive. Granite Heron may rely on retained activity and testing limits while disputing causation. Admission confirms the documents and reported observations; it does not establish that the quality decision was legally compelled or that every claimed loss followed from it.

Sample plan and assay controls

Quality personnel drew identified samples from the quarantined lot under a controlled plan. The plan retained untouched units, documented chain of possession, and compared the tested material with reference and control preparations. The laboratory recorded method suitability, instrument checks, analyst review, and deviations. No unexplained sample substitution appears in the file.

The assay measured selected activity attributes at specified conditions. It was not designed to test every vial or recreate the exact thermal history of each location within the sealed shippers. The summary therefore reports observations for the tested samples and expressly limits extrapolation. Packaging inspection found no broken exterior seal or visible container failure.

Reference responses met the method's acceptance conditions. Tested BE-214 samples retained measurable activity, although results showed dispersion requiring quality evaluation. The record does not describe the lot as universally inert. It also does not state that retained activity alone satisfies the validation protocol, which required documented process control and acceptable custody evidence. Both parties may examine whether the sample design materially informs suitability and whether further testing could have reduced uncertainty.

Reported results and uncertainty

The laboratory summary groups results by controlled sample identifier rather than by an advocate's conclusion. Values remained detectable across the tested units. Some observations were lower than the reference pattern, but the limited design did not permit a reliable lot-wide degradation percentage. No analyst certified complete loss of potency.

Blue Ember's reviewers considered the measured activity favorable information because it narrowed the concern from obvious universal destruction. They nevertheless identified unresolved questions about uniformity, transport-history compliance, and whether a validation campaign using the lot could demonstrate repeatable controlled performance. The intended use required more than proof that selected samples still responded.

Granite Heron's review comment emphasized that the assay weakened any claim of total physical loss and supported evaluation of alternate research use. Blue Ember's response agreed that limited nonvalidation investigation might be possible but stated that such use would not accomplish the planned campaign. The exchange is preserved in the file so disagreement is not hidden. Neither comment alters the logger readings, the specified band, or the carrier's lack of product-release authority.

Specification assessment

Release Specification BE214-RS-03, Revision 3 treated the transport band as a defined validation attribute and supplied no fifty-four-minute grace period. The assessment compared the 13:08 threshold crossing, minus 42.8 maximum at 13:37, and 14:02 recovery with that criterion. Reviewers also considered packaging, storage after receipt, available stability knowledge, assay results, and intended protocol use.

The assessment distinguishes a specification deviation from a scientific statement that all material was destroyed. A passing confirmatory sample could not retroactively document continuous compliant carriage. Conversely, the temperature record alone did not quantify the activity of every vial. The quality decision had to address whether the combined uncertainty was acceptable for the particular validation purpose.

The review team concluded that no existing study bridged this magnitude and duration to uniform validation suitability. It declined to create a new acceptance range after the event. Granite Heron disputed whether that caution should translate into complete economic loss and requested supporting information. The quality file records the dispute without treating the carrier's position as permission to use the lot or Blue Ember's assessment as an adjudication of damages.

September 16 disposition

On September 16, Blue Ember's authorized quality reviewers rejected BE214-VL-2409A for its planned validation campaign. Dr. Mira Solis acknowledged the program consequence but did not replace the quality approvers. The disposition cites the continuous-temperature requirement, monitor curve, limited assay, absence of a supporting excursion allowance, and the need for defensible validation custody.

The operative conclusion is "not suitable for the intended validation use." It is not "every vial inactive," "unsafe for every conceivable purpose," or "automatically worth 4.2 million dollars." Segregation continued while investigation samples, potential limited research use, claim materials, and controlled destruction were addressed. No unit was released for the planned campaign.

The document directed replacement planning, retention of representative samples, preservation of source records, and evaluation of any legitimate credit. It did not tell Granite Heron that a later manufacturing option was impossible, nor did it decide whether Blue Ember's response to such an option would be reasonable. The quality determination and subsequent mitigation dispute occupy different points in the chronology.

Accountability, residual use, and certification

The accountability schedule reconciles units received, samples withdrawn, units retained during investigation, and material sent through the controlled disposition process. Limited analytical use generated no sale or validation credit. Blue Ember identified no quantifiable salvage value that would duplicate recovery of the lot's embodied cost. Granite Heron reserved the right to challenge that conclusion through testimony and financial proof.

The custodian traced the exhibit to ordinary quality records and confirmed that later litigation summaries did not replace the September entries. Technical terms were generalized for the fictional exercise, but dates, sample relationships, monitor identifiers, and the stated basis for disposition remain fixed. Any correction is separately visible.

This exhibit may support arguments about intended use, causation, reasonableness, and damages. It may not be described as raw laboratory data interpreted by the application, proof of universal destruction, or a regulatory ruling. The court admitted the documentary presentation subject to testimony about method limits and weight. Every laboratory, material, test, and decision reflected here is synthetic.